

Linear Mixed Effects Models – Presentation Guidelines

The first three meetings of this section focus on general aspects of the application and interpretation of linear mixed effects models. For the **final meeting (March 31st)**, we'll spend time discussing specific examples of how these models are used.

To prepare for this, please find an **example of how linear mixed effects models are used in your field / area of interest**. This will likely be a peer-reviewed publication, but other formats (preprint, case study, policy brief, etc.) may work as well.

In class, you'll present this example. You should prepare a couple slides:

- **Introducing the topic and the hypothesis** / hypotheses (for a generalist audience)
- Walking through the **experimental design / data structure**
- Discussing **how the data was analyzed** with a mixed effects model (what were the fixed and random effects, why were those specific random effects chosen, how did the model structure match the hypothesis, etc.)
- Discussing **how the model was interpreted** (i.e. – how did the authors pull effect sizes, marginal means / trends, etc. from the model)

Finally, you should also **spend some time evaluating the approach**. You can discuss why the approach did or did not work, speculate on why certain analytic decisions were made, propose alternative model structures or tests, etc. These presentations are not meant to be focused on the results per se, but on **how linear mixed effects models contribute to those results**.

You should aim for a **10-12 minute presentation, with time for questions afterwards**. If you would like to prepare discussion questions for the group to go through after the presentation, even better!

To do:

Find your example study and prepare your slides.

Upload your slides to this OneDrive folder **by noon on March 31st**: [Special Problems - LME Model Presentations](#)

- Slides should be in .pdf or .pptx format
- We'll open all the slides on just one computer to avoid having to switch out computers each time